MariaDB Server Installation

> **Important**
>
> Every command on the server needs to be executed on `root` user!

1️⃣ Install MariaDB Server

1️⃣.1️⃣ Update packages

```
apt update && apt upgrade -y
```

1️⃣.2️⃣ Install MariaDB

```
apt install -y mariadb-server
```

1️⃣.3️⃣ Start and enable the service

```
systemctl start mariadb
systemctl enable mariadb
```

1️⃣.4️⃣ Verify the service is running

```
systemctl status mariadb
```

2️⃣ Secure the Installation

```
mariadb-secure-installation
```

Follow the prompts:

- Set a root password
- Remove anonymous users → **Yes**
- Disallow root login remotely → **Yes**
- Remove test database → **Yes**
- Reload privilege tables → **Yes**

3️⃣ Configure Remote Access

By default, MariaDB only listens on `127.0.0.1`. Change this to allow remote connections

3️⃣.1️⃣ Edit the configuration file

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```
vi /etc/mysql/mariadb.conf.d/50-server.cnf
```

Find the following line:

```
bind-address = 127.0.0.1
```

and change it to:

```
bind-address = 0.0.0.0
```

3️⃣.2️⃣ Restart MariaDB

```
systemctl restart mariadb
```

4️⃣ Create Remote User

Log into MariaDB as root:

```
mariadb -u root -p
```

Create a user and grant privileges (on all databases!)

```
GRANT ALL PRIVILEGES ON *.* TO 'admin'@'10.0.5.%' IDENTIFIED BY 'einfach';
FLUSH PRIVILEGES;
```

> **ⓘ Note**
>
> Replace `10.0.5.%` with `%` to allow connections from any IP, or a specific IP like `10.0.5.30`
>
> `admin` is the username and `einfach` is the password!

5️⃣ Add DNS Record

Add an `A` record to the `bind9` zone file

5️⃣.1️⃣ Edit the zone file

```
vi /etc/bind/db.3bhits.at
```

5️⃣.2️⃣ Add `A` record

Increment the serial number (e.g. from `4` to `5`):

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```
4           ; serial number
```

to:

```
5           ; serial number
```

> **Tip**
>
> You need to increase the number, so the DNS knows that the zone file has changed!

Add the new A record below the existing ones:

```
mariadb    IN      A       10.0.5.20
```

> **Important**
>
> Use the IP of your webserver!

5️⃣.3️⃣ Reload `bind9`

```
systemctl reload bind9
```

6️⃣ Test on the client

6️⃣.1️⃣ Verify DNS resolution

```
nslookup mariadb.3bhits.at
```

Expected output: `10.0.5.20` (or the IP of your webserver)

6️⃣.2️⃣ Test MariaDB connection

Install MariaDB client:

```
apt install mariadb-client -y
```

Test the connection:

```
mariadb -h mariadb.3bhits.at -u admin -p
```

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